A Lightweight and Robust Lie Group-Convolutional Neural Networks Joint Representation for Remote Sensing Scene Classification

Chengjun Xu, Guobin Zhu, and Jingqian Shu

***Abstract*— The existing convolutional neural network (CNN) models have shown excellent performance in remote sensing scene classification. However, the structure of such models is becoming more and more complex, and the learning of low-level features is difficult to interpret. To address this problem, in this study, we introduce lie group machine learning into the CNN model, try to combine both approaches to extract more distinguishing ability and effective features, and propose a novel network model, namely, the lie group regional influence network (LGRIN). First, manifold space samples of the lie group are obtained by mapping, and then, the features of the lie group are extracted after the operations of image decomposition and integral image calculation. Second, the multidilation pooling is integrated into the CNN architecture. At the same time, the image regional influence network module is designed to guide the attention of the classification model by using the regional-level supervision of the decomposition. Finally, the fusion features are classified, and the predicted results are obtained. Our model takes full advantage of regional influence, lie group kernel function, and lie group feature learning. Moreover, our model produces satisfactory performance on three public and challenging data sets: Aerial Image Dataset (AID), UC Merced, and NWPU-RESISC45. The experimental results verify that, compared with the state-of-the-art methods, this method is more explanatory and achieves higher accuracy.**

***Index Terms*— Convolutional neural network (CNN), joint representation, lie group, lie group region covariance, remote sensing scene classification.**

I. Introduction

BENEFITTING from the rapid development of various sensors and remote sensing technologies, high-resolution remote sensing (HRRS) images are widely used in the remote sensing community. Detailed spatial information can enhance the recognition of ground objects and improve classification accuracy. This trend of such development promotes the HRRS image data as useful information for various applications, such as environmental protection [1] and agriculture [2], [3]. At the same time, a new demand has been raised: whether land use and coverage categories can be automatically identified from HRRS images. Due to the need for remote sensing applications, it is particularly important to distinguish different scenes effectively. In the past long time, HRRS images have played an important role in the wide application of land use and cover classification [4], [5]. Hence, the classification of the HRRS image has attracted the attention of more and more scholars and has become a hot topic in the remote sensing community.

To obtain high accuracy classification results, effective and discriminative feature representation plays a very important role. The traditional methods are completely dependent on the spatial and low-level spectral features of HRRS images, such as scale-invariant feature transformation (SIFT) [6], the histogram of oriented gradients (HOG) [7], gray-level cooccurrence matrix (GLCM) [8], and the object spectral index [9]. These characteristics are not necessarily independent of each other, and some methods involve two or more of them. Luo *et al.* [14] captured six different types of features, namely Gabor filter, GLCM [10], [11], Gaussian wavelet [12], shape feature [13], simple radiation feature, and SIFT feature, and indexed remote sensing images according to different combinations of the above features [14]. Avramovid and Risojevid combine Gist and Spin-Echo Fourier Transform (SEFT) features to describe aerial image scenes [15]. Although directly using the low-level spectral features can better identify the scene with the spatial distribution and structural distribution, it is helpless for the scene with the complex and changeable spatial distribution of HRRS image [16]. In 2002, Plaza *et al.* [17] proposed a new automated method based on mathematical morphology, which used spatial information and spectral information to realize the interpretation and classification of remote sensing multidimensional images. In 2005, Plaza *et al.* [18] considered the spatial information and spectral information of the hyperspectral data set simultaneously to characterize the extended morphology and applied it to agricultural and urban classification problems. However, this kind of method is based on morphological vector operation and has limitations for matrix-vector space.

To address the above problems, scene classification methods based on visual the bag-of-visual-words (BoVW) model and the probabilistic topic model (PTM) have attracted the attention of scholars. The low-level features are extracted from the BoVW model, such as texture features based on shape

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invariant and SIFT features, and combined with K-means clustering, the low-level features are transformed into visual words. However, the BoVW model regards the remote sensing image as an unordered set of features and does not consider the spatial association between features [19]. To solve the lack of spatial association information, scholars proposed a method derived from the BoVW model [20]–[25]. Yang and Newsam [20], [21] calculated the cooccurrence of visual words for "spatial predicates" in the hierarchical spatial division of images. Zhao *et al.* [22] utilized standard deviation value, mean value, and dense SIFT as the scene descriptors instead of the visual word histogram classification in the BoVW model. The scene is represented as a random mixture of topics based on the PTM model method [26]–[28]. Zhong *et al.* [26] proposed a fully sparse semantic topic model, which represented images with sparse but representative semantics. Zhao *et al.* [27] proposed a PTM framework integrating local spectral features, global texture features, and local structural features.

The above methods are mainly based on handcrafted feature representations and heavily depend on prior experience information [29], [30]. To address this problem, scholars have proposed a scene classification method based on deep learning [31]–[33]. Deep learning is an "end to end" mode, which is composed of simple but nonlinear modules. The hierarchical mechanism is used to extract features, and the extracted features are transformed into more abstract and higher level features [34]. Deep learning has been widely used in image retrieval [35], speech recognition [36], [37], and object detection [38], [39]. After successfully applied to natural images, deep learning has now been introduced into the remote sensing field, such as scene classification [40]–[47] and hyperspectral pixel classification [48]–[52]. In the field of remote sensing scene classification, a deep learning method can capture features directly from the raw images without prior experience information. Zhang *et al.* [45] proposed a gradient boosting random convolution network (GBRCN) combined with different depth convolution neural networks for scene classification. Li *et al.* [53] proposed a deep feature fusion network for remote sensing scene classification. Castelluccion *et al.* [54], Hu *et al.* [41], and Wang *et al.* [40] utilized the generality of the internal layers of CNN [55], [56] to transform the CNN trained with other big data sets (such as Imagenet and Places205) into remote sensing scene classification. Jiao *et al.* [57] proposed a deep multiscale spatial–spectral feature extraction algorithm that focuses on learning effective discriminant features of hyperspectral image classification (HSIC).

Although deep learning methods have better competitive performance than handcrafted feature-based methods, they still have some common limitations.

The first limitation is that some models utilize local region features [58], [59] and middle-level layer information [60] features to describe image features, while others utilize fusion of the local and global information feature as the final representation [61]–[65]. However, they did not consider the relationship between different regions and the relationship between different features of the image. In addition to the features and object attributes themselves, the internal relationship between them is also quite important for distinguishing representations.

The second limitation is that most of the existing CNN methods only use the output of the final connection layer of the network model as the global feature representation to confirm the category of the sample [66]–[68], ignoring the local region information of the image. Two remote sensing images with different global characteristics may belong to the same category because they contain some common obvious objects. In this case, an approach that relies solely on global functionality may result in incorrect classification.

The third limitation is that most of the existing CNN models have numerous parameters and become very complex, and it is difficult to interpret the low-level feature learning. In addition, most of the existing models cannot extract more distinctive and effective features for complex spatial patterns and geometrical structures.

To address the problems of the above methods, we focus on constructing the CNN model of multifeature joint representation.

1) We utilize the image-based decomposition representation region to construct the spatial information of the image. Besides the local and global features, the relationship between different global features and the association information between different local features are encoded into the representation. These correlations are realized by using the lie group region covariance matrix.
2) The lie group samples represented by the lie group region covariance matrix do not belong to the Euclidean space and cannot be distinguished by the Euclidean distance. Motivated by the successful application of lie group machine learning and lie group kernel function in the field of remote sensing scene classification [69], [70], we try to use manifold space distance and geodesic to distinguish the differences between samples and use lie group kernel function to further improve the accuracy of classification.
3) Our model enhances the interpretability of the model from the perspective of lie group feature learning. The joint representation of multiple features not only considers the low-level features but also takes into account the high-level semantic features. In addition, the regional covariance of a lie group is a real symmetric matrix with good computational performance and antinoise ability. For the complex geometric structure and spatial pattern of the image, it has a good ability to distinguish.

This article is framed in different sections. Section II, introduces the proposed method. The experimental results and analysis are shown in Section III. The conclusion of our article with discussion is presented in Section IV.

II. Proposed Method

In this section, we will introduce our remote sensing scene classification model. First, the samples are mapped to the manifold space of the lie group to obtain the lie group samples. Then, the lie group samples are divided into regions, and the lie group features are extracted and represented by the lie

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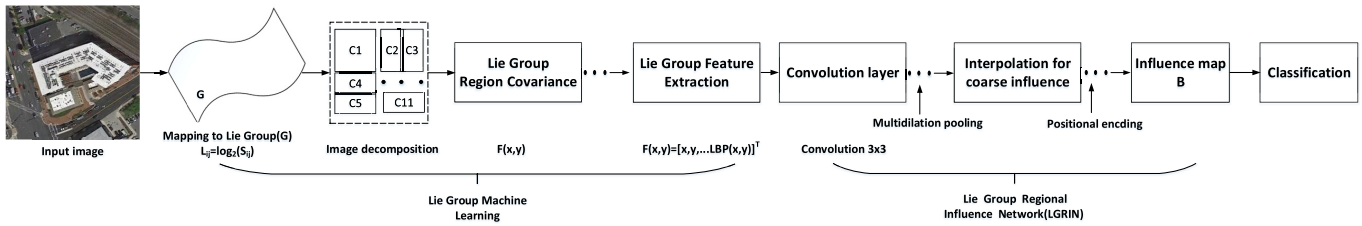


Fig. 1. General architecture of our network. The model is mainly composed of two parts. The first part is the lie group machine learning, mainly completing the regional division of lie group samples, the construction of lie group feature region covariance, and so on. The second part is the regional influence network of lie group, mainly completing convolution computation, multiscale features learning, and so on (the image is from the NWPU-RESISC45 data set).

TABLE I
SUMMARIZE THE MODEL TOPOLOGY OF OUR PROPOSED APPROACH

Input image	
Lie Group Machine Learning	Sample mapping
	Image decomposition
	Region covariance(G)
	Calculating the regional covariance distance of G
	Integral image calculation
	Feature extraction(G)
Lie Group Regional Influence Network	conv 3×3
	Multidilation pooling
	Interpolation
	conv.1D layers
	pos.injection
	Interpolation
FC	

group regional covariance. Finally, a novel lie group regional influence network (LGRIN) model with a multidilation pooling module is proposed for remote sensing scene prediction, as shown in Fig. 1. Table I lists the topology of the model.

A. Lie Group Machine Learning

1) Sample Mapping: First, we map the data samples onto the lie group manifold space to obtain the lie group samples

$$\mathbf{L}_{ij} = \log_2(S_{ij}) \tag{1}$$

where S_{ij} represents the jth sample of the ith class in the data set, and $\mathbf{L}_{ij}$ represents the jth sample of the ith class in the manifold space of the lie group. The following operations are based on the lie group data sample $\mathbf{L}_{ij}$.

2) Remote Sensing Image Decomposition: When we recognize something, we generally perceive it from the global structure. To distinguish different things, we need to further observe the local area (subregion) of things, that is, to find the difference between each subregion. Therefore, the decomposition of images conforms to the law of the development of human cognition, that is, the cognitive process from coarse to fine. At the same time, it can also quickly capture the different features of the target things. Tuzel *et al.* [71] divided the image into five regions in face detection and recognition: global region and local region. The global region captures the global image feature, and the local region is composed of four subregions, extracting features, and extracting feature vectors of nine dimensions for each pixel, respectively. Bouteldja *et al.* [86] divided the image into seventeen regions

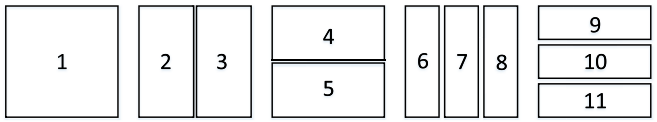


Fig. 2. Image decomposition. We construct eleven covariance matrices from the HRRS image. The covariances are used as the HRRS image descriptors.

and then designed the regional feature vectors to realize regional representation. The feature vectors of four dimensions are extracted from each pixel of the image.

Inspired by the above approaches, we conducted extensive experiments and analyses on various possible image decomposition. We have tried the diagonal method (the main diagonals and the subdiagonals) and also tried different image decomposition methods, such as overlapping decomposition and nested decomposition. The experimental results obtained by different partition methods are not the same. For example, the overall average accuracy (AA) of the experiment is low when the diagonal method is used. The river and forest scenes are easy to be confused. After an in-depth analysis, we found that there are many trees on both sides of the river, which makes the model learn more about the features of the lie group of trees, which leads to the increase in the error rate. When we divided remote sensing images into seventeen or more regions, the accuracy of some scene categories (i.e., mobile_home_park and parking_lot) is improved, and the error rate of some scenes (i.e., airplane, storage_tank) is also improved, while the overall AA of the sample did not improve. Moreover, too many regions will lead to a decrease in computing efficiency. For example, the computing time of seventeen regions is 630 ms longer than that of eleven regions. If the remote sensing image is divided into five regions, the calculation time is 710 ms faster, but the AA rate is reduced by 29.37%. Finally, to achieve a better tradeoff between AA, computing efficiency, space complexity, and time complexity, we chose the image decomposition method in Fig. 2. Thus, it can be seen that different ways of division and the results are not the same. In Section III-E, we will perform visualization and analysis of our region division to prove the effectiveness of our region division.

3) Region Covariance of Lie Groups: The covariance of the lie group features extracted from the pixels of remote sensing image represents the remote sensing image region, which is called the lie group regional covariance matrix. Use F to

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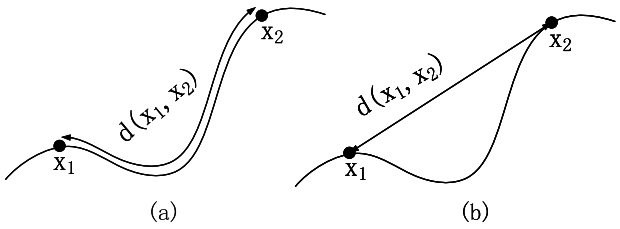


Fig. 3. Difference between the Manifold distance and the Euclidean distance. (a) Distance between the samples x_1 and x_2 on the manifold space of the lie group, and the distance is on the manifold space. (b) Distance obtained by directly using the Euclidean distance to calculate samples x_1 and x_2. The distance is not in the manifold space.

represent the features of the image extracted from image I

$$F(x, y) = \phi(I, x, y) \tag{2}$$

where ϕ represents a mapping function that extracts a d-dimensional lie group feature vector, such as the intensity, color, gradient, and filter response. For a given rectangular region $R \subset F$, the corresponding ϕ of all the pixels in the region can be represented by $\{\mathbf{z}_k\}_{k=1,2,\ldots,n}$. $\mathbf{z}_k$ represents the actual vector of d-dimensional; n represents the number of pixels within R. The region R with $d \times d$ dimensional covariance within feature points

$$\mathbf{C}_R = \frac{1}{n-1}\sum_{k=1}^{n}(\mathbf{z}_k - \boldsymbol{\mu})(\mathbf{z}_k - \boldsymbol{\mu})^{\mathrm{T}} \tag{3}$$

where $\boldsymbol{\mu}$ represents the intrinsic mean of the regional feature points. The calculation method of intrinsic mean of lie groups can be referred to [69] and [70].

There are several advantages to using a region covariance matrix as a region descriptor, and extracting a covariance matrix from a region is usually sufficient to match the region sat out under different patterns and views.

The regional covariance matrix is a method of combining multiple possible related features. The diagonal entries of the covariance matrix represent the variance of each feature, and the nondiagonal item represents the attribute correlation. In the process of covariance calculation, the noise of an individual sample is filtered on average, which reduces the impact of noise. Compared with other descriptors, the feature dimension of the covariance matrix has a lower feature dimension, and it is not affected by the size. This matrix is a real symmetric matrix, and $\mathbf{C}_R$ has only different $(d^2 + d)/2$ values, which has a better computational performance. Other detailed information about regional covariance can be referred to [71].

4) Lie Group Regional Covariance Matrix Distance Calculation: The Euclidean space is not closed when you multiply using a negative scaler. The lie group regional covariance matrix does not lie on the Euclidean space. Most of the existing machine learning models are calculated using the Euclidean space distance. However, the Euclidean space distance is not suitable for us. Fig. 3(a) shows the manifold distance, the distance between two data points in the manifold space is located in the manifold space, Fig. 3(b) directly uses the Euclidean distance to calculate the distance between two data points, which is obviously not located in the manifold space. We use the distance measure proposed in [72] to measure the distance between two covariance matrices

$$\rho(\mathbf{C}_1, \mathbf{C}_2) = \sqrt{\sum_{i=1}^{n} \ln^2 \lambda_i(\mathbf{C}_1, \mathbf{C}_2)} \tag{4}$$

where $\{\lambda_i(\mathbf{C}_1, \mathbf{C}_2)\}_{i=1,2,\ldots,n}$ represents the generalized feature values of $\mathbf{C}_1$ and $\mathbf{C}_2$, computed from

$$\lambda_i \mathbf{C}_1 x_i - \mathbf{C}_2 x_i = 0, \quad i = 1, 2, \ldots, d \tag{5}$$

and $x_i \neq 0$ are the generalized features. Equation The distance measurement measure ρ in (4) satisfies the measurement axioms of the positive symmetric matrix $\mathbf{C}_1$ and $\mathbf{C}_2$.

1) $\rho(\mathbf{C}_1, \mathbf{C}_2) \geq 0$ and $\rho(\mathbf{C}_1, \mathbf{C}_2) = 0$ only if $\mathbf{C}_1 = \mathbf{C}_2$.
2) $\rho(\mathbf{C}_1, \mathbf{C}_2) = \rho(\mathbf{C}_2, \mathbf{C}_1)$.
3) $\rho(\mathbf{C}_1, \mathbf{C}_2) + \rho(\mathbf{C}_1, \mathbf{C}_3) \geq \rho(\mathbf{C}_2, \mathbf{C}_3)$.

The distance measure also starts from the lie group structure of the positive definite matrix and derives an equivalent form from lie algebra of positive definite matrix. The literature [73] gives that any two positive definite matrices $\mathbf{S}_1, \mathbf{S}_2 \in \mathbf{Sym}^+(n)$ remain radiative invariance

$$d(\mathbf{S}_1, \mathbf{S}_2) = \|\log_2\left(\mathbf{S}_1^{-\frac{1}{2}}\mathbf{S}_2\mathbf{S}_1^{-\frac{1}{2}}\right)\|. \tag{6}$$

After adding the operation $\mathbf{S}_1 \bigodot \mathbf{S}_2 = \exp(\log_2(\mathbf{S}_1) + \log_2(\mathbf{S}_2))$, the positive symmetric group constitutes the lie group, and the distance between the two elements can be used

$$d(\mathbf{S}_1, \mathbf{S}_2) = \|\log_2(\mathbf{S}_2) - \log_2(\mathbf{S}_1)\|. \tag{7}$$

The new intrinsic mean calculation equation of $\mathbf{Sym}^+(n)$ group can be obtained

$$\boldsymbol{\mu} = \exp\left(\frac{1}{n}\sum_{i=1}^{n}\log_2(x_i)\right). \tag{8}$$

Equation (8) is called the log-Euclidean intrinsic mean. Its characteristics are given as follows.

1) The log-Euclidean intrinsic mean is calculated as the global intrinsic mean. From a geometric point of view, it is to calculate the geometric average after mapping all lie group element samples to lie algebraic linear space, which is the intrinsic mean of the sample set directly in the flow space of the lie group.
2) As the calculation of the intrinsic mean is global and only needs to be calculated once, the time complexity of the algorithm is $O(n)$.

5) Integral Image Calculation: Integral image is an intermediate image representation used for fast calculation of the sum of regions [74]. Each pixel of the entire image is the sum of all the pixels in the rectangle bounded by the upper left corner of the image and the pixels of interest. For the given image I, the integral image is defined as

$$\text{Integral Image}(x', y') = \sum_{x<x', y<y'} I(x, y). \tag{9}$$

Using this representation, the sum of any rectangular region can be calculated in constant time. In [75], integral images are extended to higher dimensions and can quickly calculate

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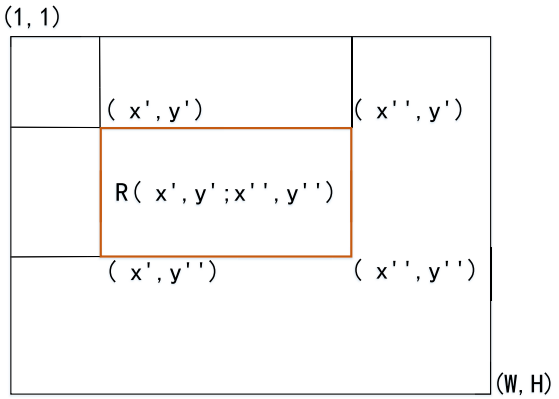


Fig. 4. Integral image. The region $(x', y'; x'', y'')$ is defined by its upper left (x', y') and lower right (x'', y'') corners in the image, and each point is a d-dimensional vector.

regional histograms. Inspired by this method, we can redefine the lie group covariance matrix defined in (10) as follows:

$$\mathbf{C}_R = \frac{1}{n-1}\sum_{k=1}^{n}(\mathbf{z}_k(i)-\boldsymbol{\mu}(i))(\mathbf{z}_k(j)-\boldsymbol{\mu}(j)). \tag{10}$$

To find the regional covariance of lie groups in a given rectangular region R in remote sensing images, we need to calculate the sum of each feature dimension, $\mathbf{z}(i)_{i=1,2,\ldots,n}$, and the multiplication of features of any two dimensions, $\mathbf{z}(i)\mathbf{z}(j)_{i,j=1,2,\ldots,n}$. We construct d^2+d integral image for each feature dimension $\mathbf{z}(i)$ and multiplication of any two feature dimensions $\mathbf{z}(i)\mathbf{z}(j)$.

$\mathbf{T}$ represents $W \times H \times d$ integral image tensor

$$\mathbf{T}(x', y', i) = \sum_{x<x', y<y'} F(x, y, i), \quad i = 1, 2, \ldots, d. \tag{11}$$

$\mathbf{T}_s$ is the $W \times H \times d \times d$ tensor of the second-order integral images

$$\mathbf{T}_s(x', y', i, j) = \sum_{x<x', y<y'} F(x, y, i)F(x, y, j), \ i = 1, 2, \ldots, d. \tag{12}$$

In [74], it is pointed out that the integral image only needs to be calculated once. In this study, $\mathbf{T}_{(x,y)}$ represent the d-dimensional vector, $\mathbf{T}_{s(x,y)}$ represent the $d \times d$ matrix

$$\mathbf{T}_{x,y} = [\mathbf{T}(x, y, 1)\ldots\mathbf{T}(x, y, d)]^{\mathrm{T}}$$
$$\mathbf{T}_{s(x,y)} = \begin{bmatrix} \mathbf{T}_s(x, y, 1, 1) & \cdots & \mathbf{T}_s(x, y, 1, d) \\ \vdots & \ddots & \vdots \\ \mathbf{T}_s(x, y, d, 1) & \cdots & \mathbf{T}_s(x, y, d, d) \end{bmatrix}. \tag{13}$$

$\mathbf{T}_{s(x,y)}$ are symmetric matrices, and just need $d+(d^2+d)/2$ to figure out $\mathbf{T}$ and $\mathbf{T}_s$. The computational complexity of constructing integral image is $O(d^2WH)$. $R(x', y'; x'', y'')$ is the rectangular region, (x', y') represents the upper left coordinate, and (x'', y'') represents the lower right coordinate, as shown in Fig. 4.

The lie group regional covariance of $R(1, 1; x', y')$ is expressed as follows:

$$\mathbf{C}_R(1, 1; x', y') = \frac{1}{n-1}\left[\mathbf{T}_{s(x',y')} - \frac{1}{n}\mathbf{T}_{x',y'}\mathbf{T}_{x',y'}^{\mathrm{T}}\right] \tag{14}$$

where $n = x' \cdot y'$. Similarly, the lie group regional covariance of $R(x', y'; x'', y'')$ is expressed as

$$\begin{aligned}\mathbf{C}_R(x', y'; x'', y'') &= \frac{1}{n-1}\Big[\mathbf{T}_{s(x'',y'')} + \mathbf{T}_{s(x',y')} - \mathbf{T}_{s(x'',y')} \\ &\quad -\mathbf{T}_{s(x',y'')} - \frac{1}{n}(\mathbf{T}_{x'',y''} + \mathbf{T}_{x',y'} - \mathbf{T}_{x',y''} - \mathbf{T}_{x'',y'}) \\ &\quad \times (\mathbf{T}_{x'',y''} + \mathbf{T}_{x',y'} - \mathbf{T}_{x',y''} - \mathbf{T}_{x'',y'})^{\mathrm{T}}\Big].\end{aligned} \tag{15}$$

The calculation amount of covariance integral image of any rectangular region is $O(d^2)$. Other detailed contents of lie group theory can be found in [76]–[78].

6) Region Covariance Lie Group Feature Extraction: In general, as described in the literature [81], pixel position, RGB color feature, and gradient information are used to construct the covariance matrix features. However, for scene target classification, there are various changes, such as size and shape, which directly affects the accuracy of scene classification. The basic location information, gradient information, and RGB features have the following shortcomings: 1) they contain less information and cannot describe the size, shape, and other attributes of the scene target and 2) they are not stable and vulnerable to changes in perspective. Therefore, to address the above problems, this study proposes a novel lie group covariance feature

$$\begin{aligned}F(x, y) = \Big[&x, y, N_R, N_G, N_B, Y, C_b, C_r, \left|\frac{\partial I(x, y)}{\partial x}\right| \\ &\left|\frac{\partial I(x, y)}{\partial y}\right|, \left|\frac{\partial^2 I(x, y)}{\partial x^2}\right|, \left|\frac{\partial^2 I(x, y)}{\partial y^2}\right| \\ &\mathrm{Gabor}(x, y), \mathrm{LBP}(x, y)\Big]^{\mathrm{T}}.\end{aligned} \tag{16}$$

Pixel position and gradient are the most basic feature information of the target. The same scene will contain similar objects. Although these objects have different sizes and shapes, their positions in the scene are similar, and the change rate of the pixels is similar. To a certain extent, they ensure the position and gradient of the pixels for the whole scene, invariance of the scene target. Therefore, we first retain these two basic feature information when constructing the covariance feature of the lie group.

In [71], the RGB color is used as one of the basic features. For different scene target objects in the HRRS image, color is a feature descriptor with high discrimination, such as white clouds in sky scenes, blue sea in the beach scene, and green leaves in the forest scene. However, only relying on a single color feature is not enough. The single color feature is easily affected by the angle of view, illumination, and so on and is unstable, which cannot address the error problem caused by the perspective difference. Therefore, to address the above problems, we should fully consider the effect of color features on the stability and regionalization of scene objects, as well as the influence of perspective differences and illumination. We first normalized the RGB color and then increased the brightness, color difference, and saturation of the $YCbCr$ space. Experiment [82] showed that this method has good robustness for the variance of perspective, and

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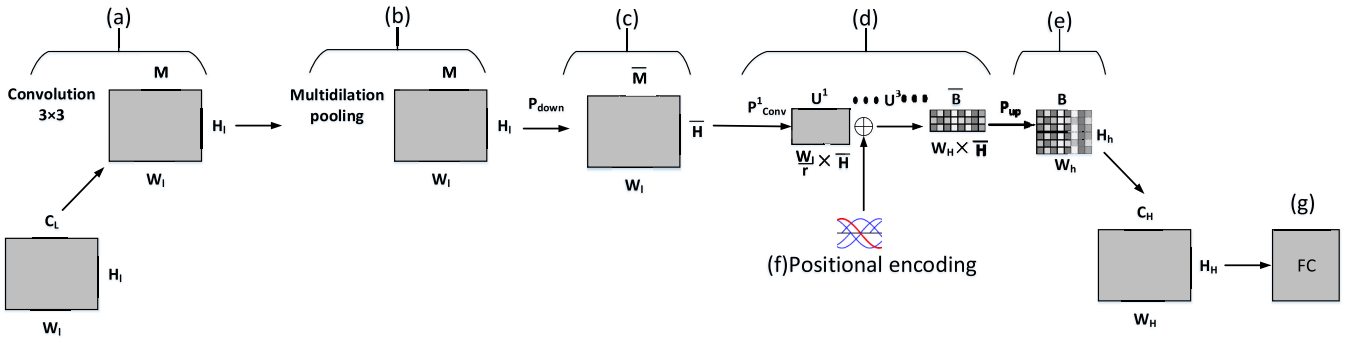


Fig. 5. Architecture of our proposed LGRIN. LGRIN mainly completes convolution computation, multiscale feature learning, position encoding, and other operations.

saturation information is less affected by illumination, with good stability. Therefore, we cascaded the information features of R, G, B, brightness, and color difference in the RGB space and $YCbCr$ space, increased the proportion of color descriptor in the covariance feature matrix, and enhanced the distinguishing ability and robust performance of scene target object classification under different conditions through the underlying feature descriptor of HRRS image

$$\begin{cases} N_R = \dfrac{R}{R+G+B} \\ N_G = \dfrac{G}{R+G+B} \\ N_B = \dfrac{B}{R+G+B} \\ Y = 0.299R + 0.587G + 0.114B \\ C_b = 0.5678(B-Y) \\ C_r = 0.713(R-Y). \end{cases} \tag{17}$$

Gabor(x, y) [83] represents the gray-scale image $f(x, y)$ of the scene, which is filtered by Gabor filter in the direction of n at the scale of m, so that the pixels of each position form the feature of $m \times n$. Its expression is given as follows:

$$\begin{cases} \text{Gabor}(x,y) = f(x,y) * g_{mn}(x,y) \\ g_{mn}(x,y) = a^{-m} g\left(x', y'\right) \\ g(x,y) = \dfrac{1}{2\pi\sigma_x\sigma_y}\exp\left[-\left(\dfrac{x^2}{\sigma^x} + \dfrac{y^2}{\sigma^y}\right)\right] \\ \qquad\qquad \times\cos(2\pi f_0 x + \varphi) \\ x' = a^{-m}(x\cos\theta + y\cos\theta) \\ y' = a^{-m}(-x\sin\theta + y\cos\theta) \\ \theta = \dfrac{n\pi}{n+1} \end{cases} \tag{18}$$

where a^{-m} represents the scale factor, σ_x and σ_y represent the standard deviation of the Gaussian function, f_0 represents the center frequency of the filter, θ represents the direction of the filter, and m and n represent the scale number and direction number of the filter, respectively. In this experiment, $m = 3$, and $n = 5$. The Gabor filter can simulate the single-cell receptive field of the cerebral cortex and capture prominent features, such as spatial positioning and direction selection [84]. For HRRS images, it can form a compact and robust representation of the middle-level semantic features of the target image. It realizes the full combination of low-level features and middle-level semantic descriptions and produces a better feature extraction effect.

Local Binary Pattern (LBP)(x, y) [85] represents the result of binarization operation on 8 pixels of the pixel field block of (x, y) and its surrounding 3×3 pixels. For the decomposed images, it can effectively describe the texture features of the scene target objects, and it is invariant to monotonous illumination changes, which enriches the spatial feature information of the lie group regional covariance descriptor and enhances the semantic description of the target image.

B. Construction of the LGRIN

To take advantage of the correlation between the low- and middle-level features of remote sensing images, we extracted fourteen different lie group features and used the lie group regional covariance to represent these features and their correlations. In this section, we further propose an LGRIN model to extract the high-level features related to the above features for remote sensing image scene classification (RESISC). LGRIN generates per-subregion scaling factors for individual partitions from its feature information as its architecture illustrated in Fig. 5.

Let $\mathbf{C}_L \in \mathbb{R}^{H_l \times W_l}$ and $\mathbf{C}_H \in \mathbb{R}^{H_h \times W_h}$ represent the feature maps of the lower level and the high level in the network model, where H and W, respectively, represent the width and height of the covariance matrix of the input lie group region. Given a low-level feature map $\mathbf{C}_L$, F_{LGRIN} generates a subregionwise influence map $\mathbf{B} \in \mathbb{R}^{H_h \times W_h}$, which is composed of the scaling factor on each subregion and fit to the high-level feature map $\mathbf{C}_H$ of higher dimensions. This process consists of the following steps: convolution layer module [see Fig. 5(a)], a multidilation pooling module for the covariance of the lie group region [see Fig. 5(b)], interpolation for coarse influence [see Fig. 5(c) and (e)], and an influence map for calculating the influence of the region [see Fig. 5(d)], in which positional encoding is added [see Fig. 5(f)]. Finally, the classification results are obtained by a fully connected (FC) layer [see Fig. 5(g)] on the concatenated the regional covariance feature matrix of the lie group.

1) Convolution Layer Module [see Fig. 5(a)]: Inspired by MobileNet V2, we choose 3×3 depthwise separable convolutions because the computation is eight to nine times less than the standard convolutions [87].

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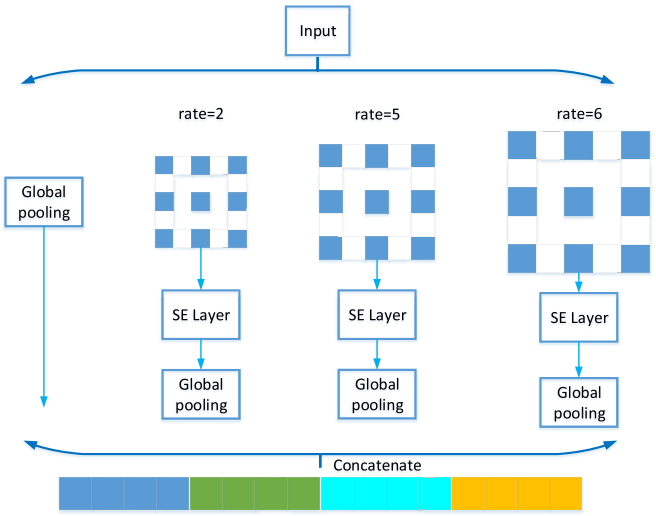


Fig. 6. Multidilation pooling module. Multiple different dilation rates are set, which can extract the features of different scales and have good robustness.

2) Multidilation Pooling Module [see Fig. 5(b)]: To capture higher level abstract features and achieve multiscale feature representation, a multidilation pooling module [88] is used in the model. Fig. 6 shows the details of the structure of this module, which is a pyramid pooling module. The idea of a pyramid pooling module comes from Stochastic Places Petri Network (SPPNet) [89]. It can be seen from Fig. 6 that this module has multiple branches, which can complete the extraction of features of different scales and has good robustness. One branch is used to directly use the global average pooling, and the other three branches are divided into global average pooling, dilated convolution, and Spatial Enhancement (SE) layer. Finally, the features of multiple branches are joined together to capture the global multidimensional feature representation. Multiple dilation rates of 2, 5, and 6 are set in three dilated convolutions to extract features of different dimensions.

3) Interpolation for Coarse Influence [see Fig. 5(c) and (e)]: After multidilation pooling operations, the model generates a matrix $\mathbf{M} \in \mathbb{R}^{W_l \times H_l}$. However, not all matrices $\mathbf{M}$ may be necessary for computing an effective influence map. As shown in Fig. 2, even if the remote sensing image is divided into 11 regions, the feature matrix of each region is quite different from the other. Therefore, we interpolate $W_l \times H_l$ matrix $\mathbf{M}$ into $W_l \times \overline{H}$ matrix $\overline{\mathbf{M}}$ by downsampling it [see Fig. 5(c)]. $\overline{H}$ is a hyperparameter, which is empirically set to 12. Since the influence map constructed by the downsampled representation is also coarse, it is converted into a dimension with the same height as the given high-level feature map $\mathbf{C}_H$ by upsampling [as shown in Fig. 5].

4) Influence Map for Calculating the Influence of the Region [see Fig. 5(d)]: The interpolation feature matrix $\overline{\mathbf{M}}$ is taken as input. Since each subregion is related to its adjacent regions, the convolution layer is selected to consider the relationship between each subregion when estimating influence mapping.

The influence map $\mathbf{B}$ indicates that regions are critical. To be able to represent multiple features, the sigmoid function is used instead of the softmax function when calculating the influence map. These operations, which consist of N convolution layers, can be written as follows:

$$\mathbf{B} = \mathbf{P}_{\text{up}}\left(\mathfrak{s}\left(\mathbf{P}^{N}_{\text{Conv}}\left(\cdots \mathfrak{R}\left(\mathbf{P}^{1}_{\text{Conv}}\left(\overline{M}\right)\right)\right)\right)\right) \tag{19}$$

where $\mathfrak{s}$ represents the sigmoid function, $\mathfrak{R}$ represents the Rectified Linear Unit (ReLU) activation, and $\mathbf{P}^{i}_{\text{Conv}}$ represents the ith 1-D convolutional layer. For the matrix, the lie group sample data set, since the matrix itself has no dot product operation, and the matrix multiplication results in the group operation are still the matrix; therefore, the vector-based kernel function cannot be used in the matrix lie group sample data set, so we use the lie group kernel function [69], [70], and the sigmoid kernel function of lie group is given as follows:

$$k(x, y) = \frac{1}{1 + e^{c\left(tr\left(y^{H} \cdot x\right)\right)}} \tag{20}$$

where tr represents the trace calculation of the matrix, c represents constant ($c < 0$), and y^H is the conjugate transpose matrix of $\overline{y}^{\text{T}}$

$$y^{H} = \begin{bmatrix} \overline{a}_{11} & \overline{a}_{12} & \cdots & \overline{a}_{1n} \\ \overline{a}_{21} & \overline{a}_{22} & \cdots & \overline{a}_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ \overline{a}_{n1} & \overline{a}_{n2} & \cdots & \overline{a}_{nn} \end{bmatrix} = \overline{y}^{\text{T}}. \tag{21}$$

According to the experience summarized in the experiment, we used four convolutional layers, which are, respectively, $\mathbf{P}^{1}_{\text{Conv}}(\overline{\mathbf{M}}) = \mathbf{U}^{1} \in \mathbb{R}^{(C_l)/(r) \times \overline{H}}$, $\mathbf{P}^{2}_{\text{Conv}}(\mathfrak{R}(\mathbf{U}^{1})) = \mathbf{U}^{2} \in \mathbb{R}^{2 \cdot (C_l/r) \times \overline{H}}$, $\mathbf{P}^{3}_{\text{Conv}}(\mathfrak{R}(\mathbf{U}^{2})) = \mathbf{U}^{3} \in \mathbb{R}^{3 \cdot (C_l/r) \times \overline{H}}$, and $\mathbf{P}^{4}_{\text{Conv}}(\mathfrak{R}(\mathbf{U}^{3})) = \overline{\mathbf{B}} \in \mathbb{R}^{W_h \times \overline{H}}$ for generating an influence map. Parameter r reduces the parameter overhead of the LGRIN model and gives potential regularization effects. After the extensive experiments, we set $r = 32$.

5) Positional Encoding Are Added [see Fig. 5(f)]: According to different regions (spatial position), scene images generally involve common structural priors, that is, each region (spatial position) in the image has significantly different lie group feature statistics in terms of category distribution. In other words, when humans recognize a remote sensing scene, they have *a priori* knowledge of the spatial position of specific objects, such as the position of buildings. Inspired by this observation, we added the sinusoidal positional encodings [90] to the intermediate feature map $\mathbf{U}^{i}$ at the ith layer of the model. After that, the hyperparameter i is analyzed. To inject positional coding, we apply the strategy in Transformer [90]. The dimension of positional coding is the same as that of the intermediate feature map (d). Specifically

$$pe_{(p,2i)} = \sin\left(\frac{p}{100^{\frac{2i}{d}}}\right) \tag{22}$$

$$pe_{(p,2i+1)} = \cos\left(\frac{p}{100^{\frac{2i}{d}}}\right) \tag{23}$$

where p represents the regional position index of the image and i represents the dimension. The new expression $\overline{\mathbf{U}}$ incorporating positional encodings is

$$\overline{\mathbf{U}} = \mathbf{U} \oplus pe \tag{24}$$

where $\oplus$ represents the elementwise sum.

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TABLE II
EXPERIMENTAL ENVIRONMENT PARAMETERS

Item	Content
Processor	Inter Core i7-4700 CPU with $2.70GHz \times 12$
Memory	32GB
Operating system	Windows 7 Pro
Hard disk	1T
Software	Matlab 2019a
GPU	Nvidia Titan-X $\times 2$
PyTorch	v1.1
Learning rate	10^{-8}
Momentum	0.9
Weight decay	5×10^{-4}

III. EXPERIMENTS

A. Data Sets

To verify the feasibility and effectiveness of our proposed method, we choose, UC Merced [20], Aerial Image Dataset (AID) [16], and NWPU-RESISC45 [91], three public and challenging data sets for testing. UC Merced data set [20] contains 21 categories, and each category contains 100 images, all of which are extracted from the U.S. Geological Survey (USGS) National Map Urban Area Imagery. The AID data set [16] contains 30 categories, each with 200 to 400 images. The NWPU-RESISC45 data set [91] contains 45 categories, each containing 700 images. It has the following three characteristics: 1) large remote sensing image data contained in the data set, 2) abundant remote sensing images, and 3) between-class similarity and high with-class diversity.

B. Experiment Setup

For the UC Merced data set [20], the training rate is set to 50% and 80%. For the AID data set [16], the training rate is set to 20% and 50%. For the NWPU-RESISC45 data set [91], the training rate is set to 10% and 20%. As shown in Table II, the parameters and equipment used in the experiment are listed. Since the number of images in each category is equal, the value of the overall accuracy (OA) is equal to the AA. Therefore, we just use the OA, confusion matrix, and kappa coefficient to evaluate all the classification methods. In addition, to calculate the confusion matrix, we used the best model in the training process.

C. Experiment Results

1) Result on UC-Merced Data Set: Previous studies [16], [91] have shown that the method based on the CNNs model far surpassed the method based on handcrafted features, so we did not compare it with the traditional method based on handcrafted features. As shown in Table III, when the training rate is 50%, our method can achieve 98.61%, surpassing all the previous methods. This indicates that higher classification accuracy can be achieved by regional division, the capture of lie group features, and the introduction of multidilation pooling modules. When the training ratio is 80%, the classification accuracy of our method achieves 98.97%. Pyramid pooling modules are used in [94], but our approach is 2.3% higher than theirs, indicating that multidilation pooling modules are more effective than pyramid pooling modules. Our method is 2.64% higher than MobileNet [79], 0.05% higher than LiG with sigmoid kernel [69], and 4.6% higher than EfficientNet [80]. Compared with Contourlet CNN [102], the accuracy is almost the same, only with small fluctuations. Moreover, we also analyzed the kappa coefficient corresponding to each approach. As shown in Table IV, the kappa coefficient of our method is 97.89%, higher than 0.08% of Contourlet CNN [102], 0.15% of SE-Multidilation Pooling Network (MDPMNet) [88], 2.98% of MobileNet [79], 0.26% of LiG with sigmoid kernel [69], and 5.52% of EfficientNet [80], which verifies the superiority of our method.

TABLE III
OVERALL ACCURACIES (%) OF EIGHTEEN KINDS OF METHODS AND OUR METHOD UNDER THE TRAINING RATIOS OF 50% AND 80% IN UC MERCED

Method	Training ratios	
	50%	80%
Fusion by addition [93]	–	97.42 ± 1.79
Pre-trained-AlexNet-SPP-SS [94]	–	96.67 ± 0.94
CaffeNet [16]	93.98 ± 0.67	95.02 ± 0.81
VGG-VD-16 [16]	94.14 ± 0.69	95.21 ± 1.20
GoogLeNet [16]	92.70 ± 0.60	94.31 ± 0.89
SRSCNN [95]	–	95.57
TEX-Net-LF [96]	96.91 ± 0.36	97.72 ± 0.54
Integrating multilayer features of CNN [97]	–	98.81 ± 0.38
D-CNN with AlexNet [65]	–	96.67 ± 0.10
D-CNN with GoogleNet [65]	–	97.07 ± 0.12
D-CNN with VGGNet-16 [65]	–	98.93 ± 0.10
Two-Stream Deep Fusion Framework [100]	96.97 ± 0.75	98.02 ± 1.03
Fine-tune MobileNet V2 [88]	97.88 ± 0.31	98.13 ± 0.33
SE-MDPMNet [88]	98.57 ± 0.11	98.95 ± 0.12
MobileNet [79]	92.17 ± 0.13	96.33 ± 0.15
EfficientNet [80]	90.91 ± 0.16	94.37 ± 0.14
Contourlet CNN [102]	–	98.97 ± 0.21
LiG with sigmoid kernel [69]	98.32 ± 0.13	98.92 ± 0.35
Proposed	98.61 ± 0.22	98.97 ± 0.31

TABLE IV
OVERALL ACCURACIES (%) AND THE KAPPA COEFFICIENT OF EIGHTEEN KINDS OF METHODS AND OUR METHOD UNDER THE TRAINING RATIOS OF 80% IN UC MERCED

Method	OA(80%)	Kappa(%)
Fusion by addition [93]	97.42	95.38
Pre-trained-AlexNet-SPP-SS [94]	96.67	95.14
CaffeNet [16]	95.02	94.33
VGG-VD-16 [16]	95.21	94.75
GoogLeNet [16]	94.31	92.26
SRSCNN [95]	95.57	94.91
TEX-Net-LF [96]	97.72	96.57
Integrating multilayer features of CNN [97]	98.81	97.64
D-CNN with AlexNet [65]	96.67	95.03
D-CNN with GoogleNet [65]	97.07	95.29
D-CNN with VGGNet-16 [65]	98.93	97.62
Two-Stream Deep Fusion Framework [100]	98.02	96.34
Fine-tune MobileNet V2 [88]	98.13	96.92
SE-MDPMNet [88]	98.95	97.74
MobileNet [79]	96.33	94.91
EfficientNet [80]	94.37	92.37
Contourlet CNN [102]	98.97	97.81
LiG with sigmoid kernel [69]	98.92	97.63
Proposed	98.97	97.89

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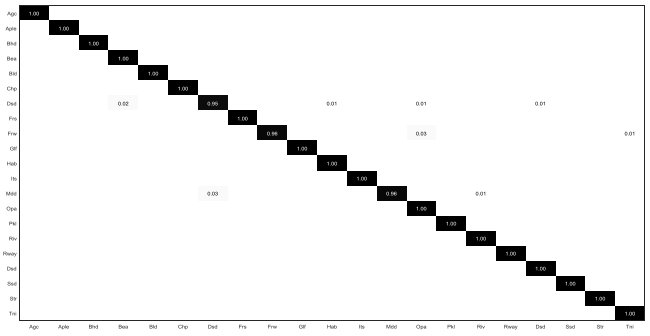


Fig. 7. Confusion matrix of our proposed method with the UC Merced data set.

As shown in Fig. 7, our method fully recognized most of the scene categories, but the recognition rate for medium residential and dense residential is lower than for other categories. Therefore, we believe that there is a large confusion between medium residential and dense residential, mainly because their distributions have great similarities, and the regional covariance of lie groups captured is less different.

2) Result on AID Data Set: The AID data set [16] consists of 30 categories and contains 10 000 remote sensing images. The remote sensing images in AID are multisource images, which is more challenging than the classification of single-source images. Compared with the UC Merced data set [20], the AID data set [16] has a smaller interclass dissimilarity but higher intraclass variations and a larger scale.

Table V shows the experimental results. When the training ratio is 20%, our method achieves 94.74%, 0.06% higher than SE-MDPMNet [88], 0.57% higher than LiG with Radial Basisi Function (RBF) kernel [70], and 2.35% higher than ResNet50 [99]. When the training ratio is 50%, our method can achieve a higher classification accuracy, achieving 97.65%. Our method is even 0.51% higher than SE-MDPMNet [88], 1.46% higher than LiG with RBF kernel [70], 0.29% higher than Contourlet CNN [102], 6.74% higher than MobileNet [79], and 9.3% higher than EfficientNet [80], which verifies that our method has stronger feature expression ability, good interpretation, and understanding ability and can capture more distinguishing features without adding any fusion strategy. As shown in Table VI, the kappa coefficient of our proposed method is 96.31%, higher than 0.12% of Contourlet CNN [102], 1.98% of LiG with RBF kernel [70], 0.44% of Super SE (SSE)-MDPMNet [88], 6.78% higher than MobileNet [79], and 9.1% higher than EfficientNet [80].

As for the confusion matrix shown in Fig. 8, our method can identify most scenes, the classification accuracy can reach 93%, and some scenes can reach 100%, such as forest and park. However, some scenarios were only 87% accurate. Further analysis, due to the similar structures, we found that there was a small difference in the lie group feature matrix between the ponds and building.

3) Results on NWPU-RESISC45 Data Set: The NWPU-RESISC45 [91] data set is a publicly available benchmark of RESISC created by Northwest Polytechnic University (NWPU). There are 45 scene categories in this data set, including 31 500 remote sensing images. The data set has three notable characteristics: 1) high within-class diversity; 2) large scale; and 3) rich image variations.

TABLE V

OVERALL ACCURACIES (%) OF EIGHTEEN KINDS OF METHODS AND OUR METHOD UNDER THE TRAINING RATIOS OF 20% AND 50% IN AID

Method	Training ratios	
	20%	50%
CaffeNet [16]	86.86 ± 0.47	89.53 ± 0.31
VGG-VD-16 [16]	86.59 ± 0.29	89.64 ± 0.36
GoogLeNet [16]	83.44 ± 0.40	86.39 ± 0.55
DenseNet121 [98]	93.76 ± 0.23	94.73 ± 0.26
DenseNet169 [98]	92.43 ± 0.36	94.17 ± 0.33
VGG19 [99]	87.73 ± 0.25	91.71 ± 0.42
ResNet50 [99]	92.39 ± 0.15	94.69 ± 0.19
InceptionV3 [99]	93.27 ± 0.17	95.07 ± 0.22
Fusion by addition [93]	–	91.87 ± 0.36
Two-Stream Deep Fusion Framework [100]	92.32 ± 0.41	94.58 ± 0.25
Two-Stage Deep Feature Fusion [101]	–	94.65 ± 0.33
TEX-Net-LF [96]	93.81 ± 0.12	95.73 ± 0.16
Fine-tune MobileNet V2 [88]	94.13 ± 0.28	95.96 ± 0.27
SE-MDPMNet [88]	94.68 ± 0.17	97.14 ± 0.15
MobileNet [79]	88.53 ± 0.17	90.91 ± 0.18
EfficientNet [80]	86.56 ± 0.17	88.35 ± 0.16
Contourlet CNN [102]	–	97.36 ± 0.45
LiG with RBF kernel [70]	94.17 ± 0.25	96.19 ± 0.28
Proposed	94.74 ± 0.23	97.65 ± 0.25

TABLE VI

OVERALL ACCURACIES (%) AND THE KAPPA COEFFICIENT OF EIGHTEEN KINDS OF METHODS AND OUR METHOD UNDER THE TRAINING RATIOS OF 50% IN AID

Method	OA(50%)	Kappa(%)
CaffeNet [16]	89.53	88.36
VGG-VD-16 [16]	89.64	88.45
GoogLeNet [16]	86.39	85.27
DenseNet121 [98]	94.73	93.38
DenseNet169 [98]	94.17	93.09
VGG19 [99]	91.71	90.06
ResNet50 [99]	94.69	93.47
InceptionV3 [99]	95.07	93.91
Fusion by addition [93]	91.87	89.34
Two-Stream Deep Fusion Framework [100]	94.58	93.34
Two-Stage Deep Feature Fusion [101]	94.65	93.41
TEX-Net-LF [96]	95.73	94.37
Fine-tune MobileNet V2 [88]	95.96	94.83
SE-MDPMNet [88]	97.14	95.87
MobileNet [79]	90.91	89.53
EfficientNet [80]	88.35	87.21
Contourlet CNN [102]	97.36	96.19
LiG with RBF kernel [70]	96.19	94.33
Proposed	97.65	96.31

Table VII shows the experimental results of our method and existing methods. When the training ratio is 10%, our method achieves 91.91%, 1.75% higher than Fine-tune MobileNet V2 [88], 1.68% higher than LiG with RBF kernel [70], and 1.72% higher than LiG with sigmoid kernel [69]. When the training ratio is 20%, our method achieves 94.43%. Our approach is 0.11% higher than SE-MDPMNET [88] for 10% training and 0.32% higher than SE-MDPMNET [88] for 20% training. The experimental results show that our method can capture more distinguishing features for rich

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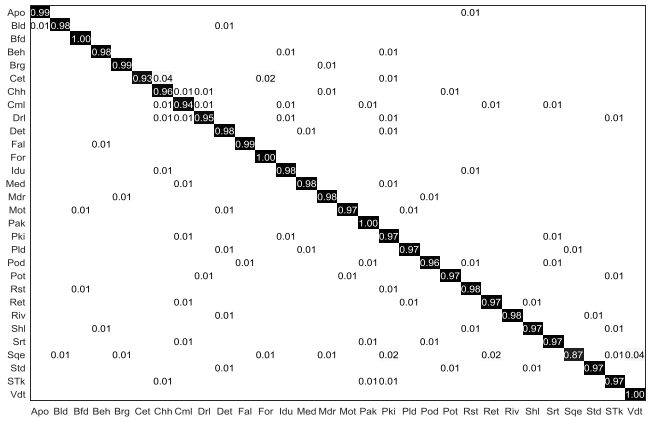


Fig. 8. Confusion matrix of our proposed method with the AID data set.

TABLE VII

OVERALL ACCURACIES (%) OF 20 KINDS OF METHODS AND OUR METHOD UNDER THE TRAINING RATIOS OF 10% AND 20% IN NWPU-RESISC45

Method	Training ratios	
	10%	20%
Fine-tune AlexNet [63]	81.22 ± 0.19	85.16 ± 0.18
Fine-tune VGGNet-16 [63]	87.15 ± 0.45	90.36 ± 0.18
Fine-tune GoogLeNet [63]	82.57 ± 0.12	86.02 ± 0.18
BoCF [63]	82.65 ± 0.31	84.32 ± 0.17
Two-Stream Deep Fusion Framework [100]	80.22 ± 0.22	83.16 ± 0.18
D-CNN with AlexNet [65]	85.56 ± 0.20	87.24 ± 0.12
D-CNN with GoogleNet [65]	86.89 ± 0.10	90.49 ± 0.15
D-CNN with VGGNet-16 [65]	89.22 ± 0.50	91.89 ± 0.22
DenseNet121 [98]	88.31 ± 0.35	90.47 ± 0.33
DenseNet169 [98]	87.93 ± 0.43	89.24 ± 0.35
VGG19 [99]	81.34 ± 0.32	83.57 ± 0.37
ResNet50 [99]	86.23 ± 0.41	88.93 ± 0.12
InceptionV3 [99]	85.46 ± 0.33	87.75 ± 0.43
Fine-tune MobileNet V2 [88]	90.16 ± 0.12	93.00 ± 0.18
SE-MDPMNet [88]	91.80 ± 0.07	94.11 ± 0.03
MobileNet [79]	80.32 ± 0.16	83.26 ± 0.17
EfficientNet [80]	78.57 ± 0.15	81.83 ± 0.15
Contourlet CNN [102]	85.93 ± 0.51	89.57 ± 0.45
LiG with sigmoid kernel [69]	90.19 ± 0.11	93.21 ± 0.12
LiG with RBF kernel [70]	90.23 ± 0.13	93.25 ± 0.12
Proposed	91.91 ± 0.15	94.43 ± 0.16

image variations, between-class similarity, and high within-class diversity. As shown in Table VIII, the kappa coefficient of our proposed method is 93.28%, which is higher than the 0.35% of Fine-tune MobileNet V2 [88], 0.32% of SE-MDPMNet [88], 11.56% of MobileNet [79], 13.75% of EfficientNet [80], 4.93% of Contourlet CNN [102], and 0.26% of LiG with RBF kernel [70].

The obfuscation matrix is shown in Fig. 9. Similar to the AID data set [16] results, the accuracy of most scenarios can reach 91%. Since the NWPU-RESISC45 data set [91] contains 45 scene categories, which contains rich image variations, has high between-class similarity, and is quite challenging, we find from Fig. 9 that none of the categories are correctly classified completely. Among them, churches and palaces are the least accurately classified. Further analysis shows that the lie group features a matrix of these two categories that are closer than those of other categories, which is easy to

TABLE VIII

OVERALL ACCURACIES (%) AND THE KAPPA COEFFICIENT OF 20 KINDS OF METHODS AND OUR METHOD UNDER THE TRAINING RATIOS OF 20% IN NWPU-RESISC45

Method	OA(20%)	Kappa(%)
Fine-tune AlexNet [63]	85.16	83.91
Fine-tune VGGNet-16 [63]	90.36	89.25
Fine-tune GoogLeNet [63]	86.02	84.76
BoCF [63]	84.32	83.24
Two-Stream Deep Fusion Framework [100]	83.16	81.94
D-CNN with AlexNet [65]	87.24	86.19
D-CNN with GoogleNet [65]	90.49	89.28
D-CNN with VGGNet-16 [65]	91.89	90.37
DenseNet121 [98]	90.47	89.33
DenseNet169 [98]	89.24	88.26
VGG19 [99]	83.57	82.17
ResNet50 [99]	88.93	87.61
InceptionV3 [99]	87.75	86.46
Fine-tune MobileNet V2 [88]	93.00	92.93
SE-MDPMNet [88]	94.11	92.96
MobileNet [79]	83.26	81.72
EfficientNet [80]	81.83	79.53
Contourlet CNN [102]	89.57	88.35
LiG with sigmoid kernel [69]	93.21	92.96
LiG with RBF kernel [70]	93.25	93.02
Proposed	94.43	93.28

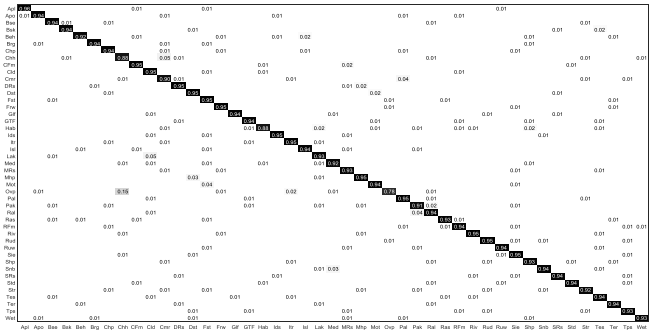


Fig. 9. Confusion matrix of our proposed method with the NWPU-RESISC45 data set.

confusion. In Section III-E, we give a detailed visualization analysis.

D. Evaluation of Size of Models

Furthermore, we choose nine typical approaches to compare the size of the model. The CaffeNet [16], GoogLeNet [16], MobileNet V2 [87], SE-MDPMNet [88], ResNet50 [99], Visual Geometry Group (VGG)-VD-16 [16], Contourlet CNN [102], LiG with RBF kernel [70], and Inception V3 [99] were used to compare with our model, and analyze the number of parameters and the size of Giga Multiply-Accumulation operations per second (GMACs). The number of parameters represents the size of the model, and the number of GMACs represents the complexity of the calculation. From Table IX, we found that our model has advantages in both computational complexity and parameters compared with Contourlet CNN [102]. Compared with the lightweight models GoogLeNet [16] and MobileNet V2 [87], our model has a better tradeoff between accuracy and model size. Despite the introduction of extended multidilation pooling modules, our model is still lightweight compared with VGG-VD-16 [16] with the extra high computation.

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TABLE IX
EVALUATION OF SIZE OF MODELS

Methods	Acc(%)	Parameters	GMACs
CaffeNet [16]	89.53	60.97M	3.6532G
GoogLeNet [16]	86.39	7M	0.7500G
MobileNet V2 [87]	95.96	3.5M	0.3451G
VGG-VD-16 [16]	89.64	138.36M	7.7500G
ResNet50 [99]	94.69	25.61M	1.8555G
SE-MDPMNet [88]	97.14	5.17M	0.9843G
Contourlet CNN [102]	89.57	12.6M	1.0583G
LiG with RBF kernel [70]	96.19	2.07M	0.2351G
Inception V3 [99]	95.07	45.37M	2.4356G
Proposed	97.65	4.63M	0.4933G

E. Qualitative Analysis

Visualization of Regional Influence: We visualize each subregion to analyze our proposed approach. The visualization of regional influence highlights the regions emphasized in different positions in the image. Through the visualization, we can find out interesting clues to verify our methods and observations. Fig. 10 clearly shows that different subregions in the image are assigned different weights, which means that our method correctly captures the most critical core areas in the image. The horizontal axis represents each subregion, and the vertical axis represents the weight, showing the weight of each subregion. The weight of each subregion is clearly different. They are clustered together for better visualization.

We extract the 14-D lie group feature vector for $C_1, C_2, \ldots, C_{11}$ and then calculate their weight. According to different regions (spatial position), scene images generally involve common structural priors, that is, each region (spatial position) in the image has significantly different lie group feature statistics in terms of category distribution. In other words, when humans identify a scene, they have prior knowledge of specific target object areas (or positions), such as the road and sky appear in the lower and upper part, respectively. Inspired by this observation, we add the sinusoidal positional encoding module [90] to the model to strengthen the weight value of lie group features of the target region (position) objects. In the process of tracking model and algorithm implementation, we decompose the image into different regions to find out the target area (position) of the object accurately. In addition to the complete aircraft, C_1 also contains other target objects (such as the runway), and it is a complete image. It does not distinguish the regions where different target objects are located, which weakens the weight proportion of aircraft to a certain extent, resulting in the low weight of the C_1 area. For C_2 and C_4, which also involve the complete aircraft, other jamming targets are much smaller than C_1. After image decomposition, the target area (position) is highlighted, the range of the target is reduced, and the target object is clearer so that the model can learn more lie group features about the aircraft. Therefore, C_2 and C_4 have more lie group features statistics compared with C_1, and the weights' value is larger. Similarly, the small difference between C_2 and C_4 is the same.

Visual Analysis of Misclassification: In the NWPU-RESISC45 data set [91], the worst results are palace and church. Therefore, we further explore the causes of misclassification. As shown in Fig. 11 and 11(a) represents the

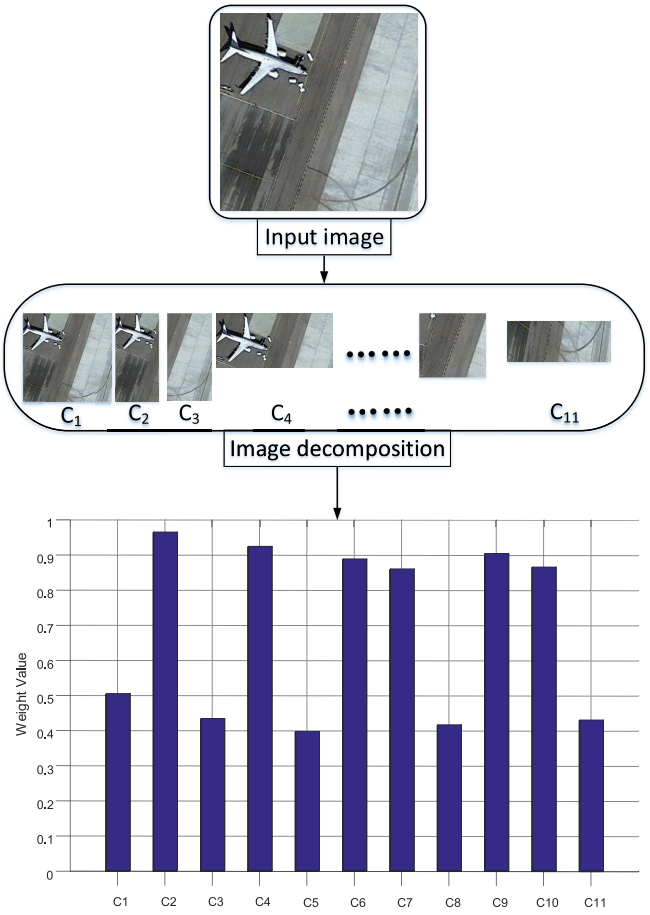


Fig. 10. Visualization of the influence of each subregion. (Upper) Input sample data. (Middle) Input sample data are divided into eleven subregions. (Lower) Visual display of the corresponding weight value of each subregion. (The image is from the NWPU-RESISC45 data set.)

visualization of the palace scene in the $X - Z$ plane, and Fig. 11(b) represents the visualization of the church scene in the $X - Z$ plane. From Fig. 11, we clearly find that palace and church have obvious high similarity. Then, we analyze the visualization of the $Y - Z$ plane, as shown in Fig. 12. Fig. 12(a) shows the visualization of the palace scene in the $Y - Z$ plane, and Fig. 12(b) shows the visualization of the church scene in the $Y - Z$ plane. From Fig. 12, we get the same conclusion.

In addition, we use (25) to calculate the degree of similarity between regions of influence

$$R = \frac{\sum_m \sum_n \left(A_{mn} - \overline{A}\right)\left(B_{mn} - \overline{B}\right)}{\sqrt{\sum_m \sum_n \left(A_{mn} - \overline{A}\right)^2 \left(B_{mn} - \overline{B}\right)^2}} \tag{25}$$

where, A_{mn} represents the $m \times n$ influence region matrix of scene A, B_{mn} represents the $m \times n$ influence region matrix of scene B, and $\overline{A}$ and $\overline{B}$ all represent the intrinsic mean of the influence region matrix. The value range of R is within $[0, 1]$, and the closer the value is to 1, the more relevant it is. Compared with the difference between other categories, the calculation results of church and palace scenes are 0.6888, which has a strong similarity. Therefore, we find that this is the root cause of its classification error.

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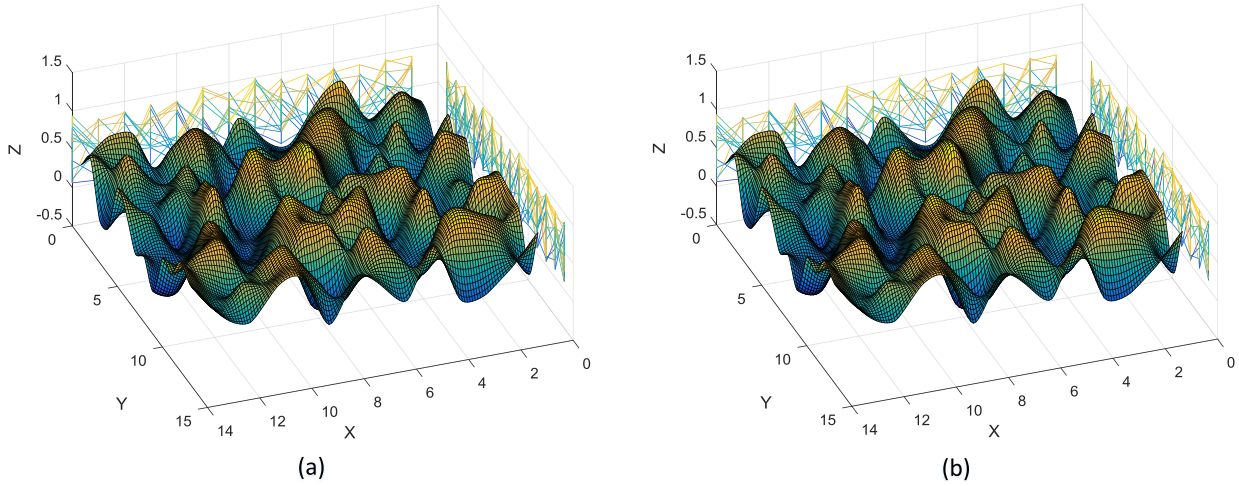


Fig. 11. Section of the $X-Z$ plane. (a) Visualization of the weights of the palace scene in the $X-Z$ plane. (b) Visualization of the weights of the church scene in the $X-Z$ plane.

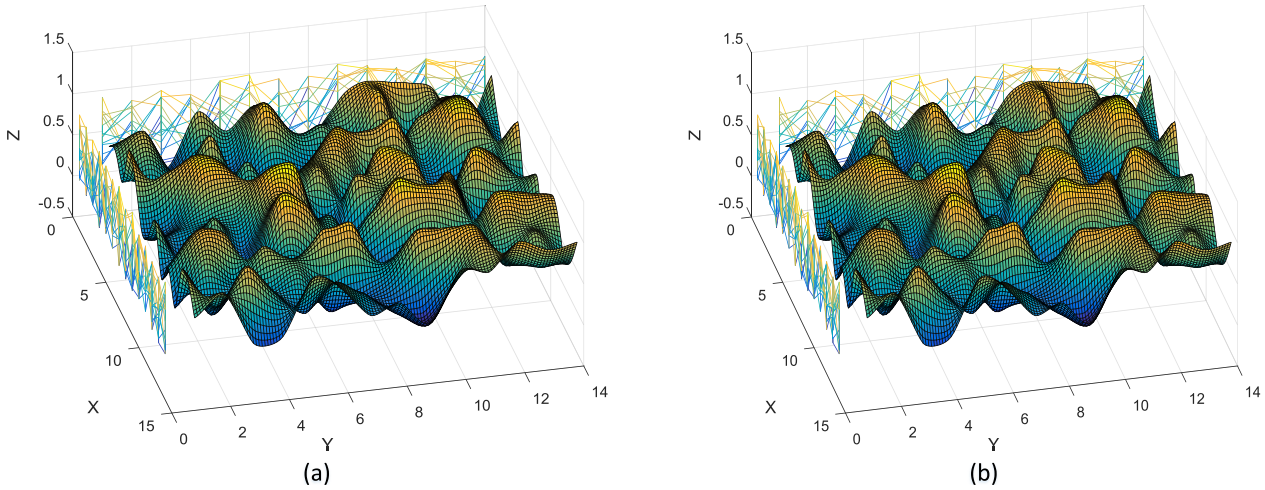


Fig. 12. Section of the $Y-Z$ plane. (a) Visualization of the weights of palace scenes in the $Y-Z$ plane. (b) Visualization of the weights of the church scenes in the $Y-Z$ plane.

IV. Conclusion

In this study, we proposed a model suitable for remote sensing scene classification, which combines the advantages of lie group regional covariance, lie group kernel function, and multidilation pooling. Our model exploits the spatial priors existing in remote sensing scene images to construct an economical and efficient architecture. By adding the lie group machine learning method to the model, we demonstrated the performance improvement without increasing the computational cost. Moreover, we visualized and analyzed the regional influence to prove the validity of our initial hypothesis, that is, the image decomposition is helpful to the classification of remote sensing images. Experiments on three publicly available and challenging data sets (UC Merced, AID, and NWPU-RESISC45) show that our method is robust and can achieve a better accuracy rate compared with the current state-of-the-art methods.

In the future, we will further study lie group machine learning and integrate it into the existing CNNs model to ensure good computing performance and improve classification accuracy.

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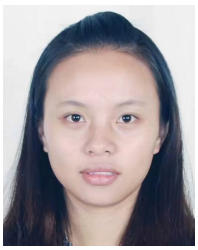
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